

ELEC4631 Lab 4

Term 2 2022

1 General information

1. This lab is conducted as both a face-to-face and **remote lab**, please follow the Remote Lab Instructions for ELEC4631 to attend it (can be found on Moodle).
2. The lab tasks must be carried out and completed during the lab time to be marked by demonstrators.
3. Lab tasks are divided into well defined time checkpoints: tasks that are scheduled to be completed in first **40 minutes**, in the first **1 hour 10 minutes**, in **1 hour 40 mins**, and in **2 hours 20 mins**. Students must pace themselves to complete the tasks and have them assessed at the defined time checkpoints.
4. Students must not arrive late at the lab by more than 10 minutes. Demonstrators can refuse to allow late students into the lab.

2 Preparatory tasks

To prepare for this lab, you will need to:

1. Review Lectures 9 and 10 on LMIs and state feedback design using LMIs.
2. Attempt Tutorial 10 Problem 5.
3. Learn the function of, and how to use, the following Matlab commands: `setlmis`, `lmivar`, `lmiterm`, and `feasp`.

3 Laboratory

3.1 Solving LMIs in Matlab

For each of the LMIs below, use the Matlab commands `setlmis`, `lmivar`, `lmiterm`, and `feasp` to find a solution of the LMIs:

1. **Assessment Exercise 1 (3 points, assessed at 40 minutes):** The strict LMI $F(x) < 0$ with $x \in \mathbb{R}^4$, and

$$F(x) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + x_1 \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix} + x_2 \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} + x_3 \begin{bmatrix} 6 & -7 \\ -7 & 6 \end{bmatrix} + x_4 \begin{bmatrix} -1 & 5 \\ 5 & -1 \end{bmatrix}.$$

Verify your solution, if one exists, check that it does satisfy the given LMI, and show it to your demonstrator.

2. **Assessment Exercise 2 (3 points, assessed at 1 hour 10 minutes):** The strict LMIs $X > 0$ and $F(X) < 0$ with $X \in \text{sym}(\mathbb{R}^{3 \times 3})$ and

$$F(X) = \begin{bmatrix} -3 & 0 & 0 \\ 1 & -4 & 1 \\ -2 & 2 & -5 \end{bmatrix} X + X \begin{bmatrix} -3 & 1 & -2 \\ 0 & -4 & 2 \\ 0 & 1 & -5 \end{bmatrix} + \begin{bmatrix} 8 & 10 & -12 \\ 10 & 14 & -16 \\ -12 & -16 & 24 \end{bmatrix}.$$

Show your solution, if one exists, check that it does satisfy the given LMI, and verify it to your demonstrator.

3.2 Linear state feedback design for flexible joint robot control with LMIs in Matlab

In this lab task you will design a linear state-feedback controller for the single stage flexible joint robot from Lab 3 using Matlab. Do the following steps:

1. Create a new Matlab script named **eqn_robot_stage1.m** to set up your state-space model or use the file of the same name that you have saved from Lab 3.
2. Use the Matlab command **ss** in your Matlab script to set up a state-space model for Stage 1 using the formulas you used in 3 Pre-lab Exercise 1. Name the system matrices in your model, A , B , C , and D . Note that individual robots can have slightly different parameter values J_{11} , J_{12} , K_{s1} , K_{t1} , B_{11} and B_{12} , so the A, B, C, D matrices will be different for each robot. In particular, the A, B, C, D matrices you will compute based on these parameters will have numerical entries different from those in Lab 3 Pre-Lab Exercise 1. **As in Lab 3, please ask your demonstrator for the parameter values of the robot that you are working on.**

Now complete the following tasks.

Assessment Exercise 3 (8 points, assessed 1 hour and 40 minutes): You want to design a linear state feedback control law $u = -Kx$ (note that in Lab 3 the state feedback gain was labelled as $K1$) for the robot such that the closed-loop system has a convergence rate μ lying in the interval $3 < \mu < 150$. To find K you can solve the following LMI in

the matrix variable $X = X^T \in \mathbb{R}^{4 \times 4}$ and $Y \in \mathbb{R}^{1 \times 4}$ (this is the content of Tutorial 10 Problem 5):

$$\begin{aligned} X &> 0, \\ XA^T - Y^T B^T + AX - BY &< -6X \\ XA^T - Y^T B^T + AX - BY &> -300X \end{aligned}$$

If X and Y are solutions to the above LMI then the gain K is given by $K = YX^{-1}$. For this assessment exercise complete the following:

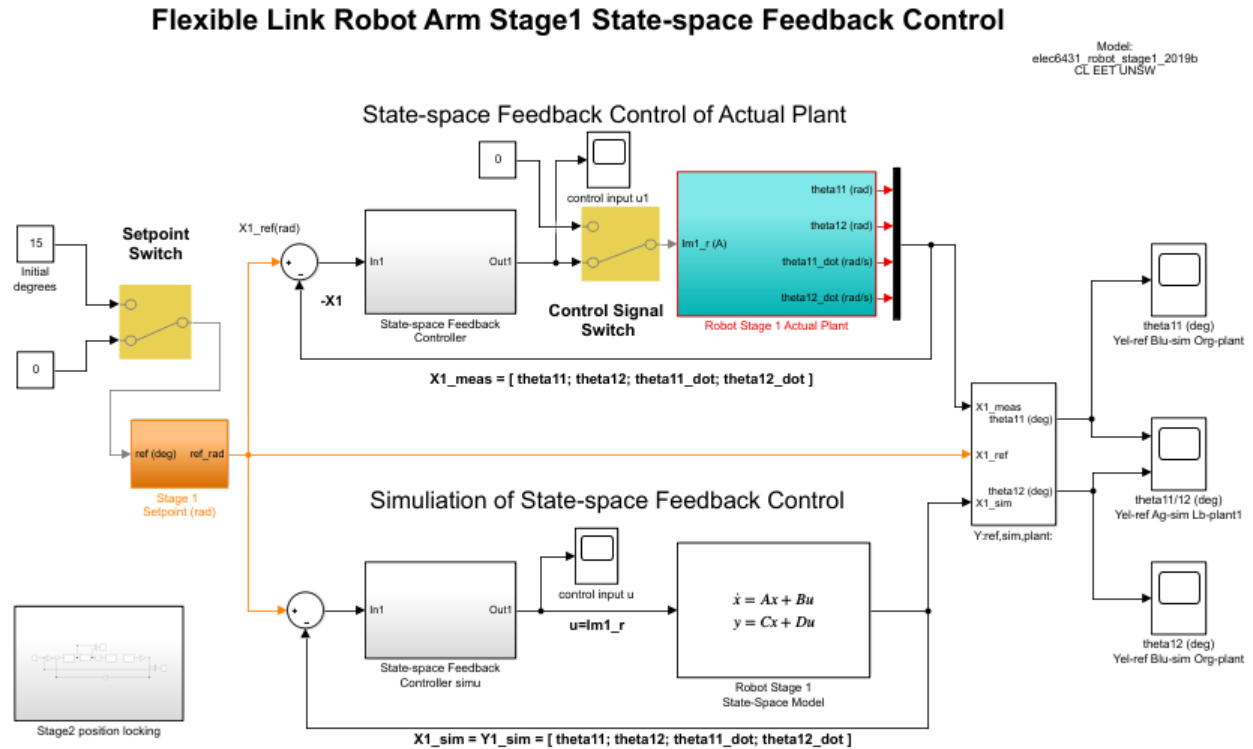
1. Add a Matlab script to solve the LMI above to obtain X , Y and K in the script **eqn_robot_stage1.m**. Show the value of X, Y, K that you have computed.
2. Show that your LMI code is correct by verifying that all the eigenvalues of the matrix $A - BK$ are in the LHP and that all their real parts lie in the interval $(-150, -3)$.

Assessment Exercise 4 (6 points, assessed at 2 hours and 20 minutes):

In this final lab exercise for the course, you will implement the linear state feedback controller that you have computed in Assessment Exercise 3 and execute it on the robot. To do this, input the value of the gain K into the Simulink model for the Stage 1 control experiment from Lab 3 into both the upper block for actual control and lower block for the simulation, as repeated in Fig. 1.

First, you must verify your controller using the simulation loop only, before applying it to the actual robot arm system. This is as in Sections 7.1.1 and 7.1.2 of Lab 3 and is repeated below.

1. Open and Run Matlab script **setup_robot_parameters.m**, located in Documents \MATLAB, to assign the physical parameters of the robot to the variables used in the system.
2. Run your script **eqn_robot_stage1.m** to define the state-space model, as well as your LMI-based controller controller, in Workspace. Double check that A , B , C , D and K are there.
3. Disable the control output to the robot in the top section: Make sure that the **Control Signal Switch** is switched to **Constant 0** (double-click it to flip over).
4. Set the setpoint to Zero: flip the **Setpoint Switch** on the left of the model to **Constant 0**.
5. Run simulation only (nothing should happen to the real robot arm): click the **Run** button. Open the scopes. Flip the **Setpoint Switch** from **constant 0** to **15**, the simulated model of Stage 1 will move away from zero and settles at the initial position to at 15 degrees. Flipping the **Setpoint Switch** to **Constant 0** will bring the angle back to 0 degree.



6. Show your simulation results to your demonstrator. And check with him that if you can start the control experiments on the real robot.

Finally, you are now ready to run the full control experiments. Before the power of robot can be switched on make sure that the control program is not running and the **Setpoint Switch** is on **Constant 0**, and **Control Signal Switch** is connected back to the robot arm. Follow the following steps:

1. Ask a demonstrator to come to your robot desk and switch on the power of the robot for you.
2. Run at Desktop Real-Time mode. The robot should not move as the Setpoint is zero. Flip the **Setpoint Switch** to **Constant -15** so the Stage 1 arm will be driven away from its central position and let the dynamics settle. This step sets the first stage to an initial position of -15° for the your LMI-based controller to reset back to the central position of 0° . Observing it on the scopes, it may take a while to return to the central position depending on how fast the system will settle. Note that the plots now show both the results of simulation (theoretical) and actual plant responses.
3. Flip the **Setpoint Switch** back to **Constant 0**, the controller will turn the Stage 1 arm back to its initial central position with transients. Record the plots observed

on the scopes in your lab notebook.

(end of elec4631 Lab 4 notes)